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AI-Based Automated Log Analysis and Performance Evaluation System for Emulation-Based Software Testing

EC481P : BE Major Project Phase 3 Presentation

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Outline

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About the organization

- Qualcomm India Pvt. Ltd. is a leading semiconductor and telecommunications company specializing in System-on-Chip (SoC) design and wireless technologies
- It develops advanced processors and communication systems used in smartphones and embedded devices
- Emulation platforms are used for pre-silicon validation of hardware and software





Introduction

- In modern semiconductor development, hardware and software are designed together (co-design) to ensure system functionality and performance.
- Emulation platforms enable testing of software even before the actual hardware is available, helping in early validation.
- Multiple software test cases are executed on these platforms, and each test generates detailed log files containing system behavior and performance data.



Introduction

- These logs include important metrics such as bandwidth, clock cycles, and instructions per second (IPS), which are essential for performance evaluation
- However, manual analysis of such large volumes of log data is time-consuming, error-prone, and inefficient, creating the need for an automated solution, The proposed system integrates LogPAI for log parsing and LogAI for machine learning-based analysis.



Motivation

- Manual analysis of a large number of test logs (around 100+ per run) is highly repetitive and consumes significant time
- Engineers spend more effort on extracting data from logs rather than focusing on root-cause analysis and problem solving
- It is difficult to quickly determine whether an issue is caused by hardware or software without proper automated comparison
- Lack of automated summarization delays communication of results to managers and cross-functional teams
- There is a strong need for efficient and faster debugging tools to meet industry timelines and improve productivity. Existing tools lack a unified hybrid AI-based approach for log analysis



Problem Statement

To overcome the slow and error-prone manual analysis of extensive log data from large-scale emulation tests, and to effectively identify performance issues and their root cause, an automated system for efficient log extraction, analysis, and summarization is required to improve debugging and decision-making.



Objectives of the Project

- To replace manual log-grepping with a single automated pipeline that runs per regression and produces repeatable outputs for any engineer in core.
- To compress regression analysis time from hours to ~2 minutes per regression by automating extraction and formatting of results.
- To standardize customer-request reporting via per-regression 3-sheet workbooks and a final multi-option comparison workbook (Cascade vs Nova-*).
- To enable fast degradation detection and root-cause direction using automatically computed % diffs and stall/bandwidth indicators (e.g., HVX stalls, DMA-bound signals).



Literature Survey

SI No	Paper Title, Journal Name, Year	Major findings / observations
1	Zhang, Y. et al. (2025) – SemanticLog: Towards Effective and Efficient Large-Scale Semantic Log Parsing(IEEE).	• LLM-based semantic parsing improves accuracy and efficiency in large-scale log analysis while maintaining privacy



Literature Survey

Sl No	Paper Title, Journal Name, Year	Major findings / observations
2	Liu, X. et al. (2025) – LUK: Empowering Log Understanding With Expert Knowledge From Large Language Models.(IEEE Transactions)	• Incorporating LLM-based expert knowledge significantly enhances log understanding and analysis performance.



Literature Survey

Sl No	Paper Title, Journal Name, Year	Major findings / observations
3	Wang, H. et al. (2025) – LogDLR: Unsupervised Cross-System Log Anomaly Detection Through Domain-Invariant Latent Representation(IEEE).	• Domain-invariant representations improve anomaly detection across different systems with varying log formats.



Literature Survey

Sl No	Paper Title, Journal Name, Year	Major findings / observations
4	Chen, Z. et al. (2025) – EPAS: Efficient Online Log Parsing via Asynchronous Scheduling. IEEE ICDE.	Asynchronous scheduling improves the speed and efficiency of log parsing in real-time systems



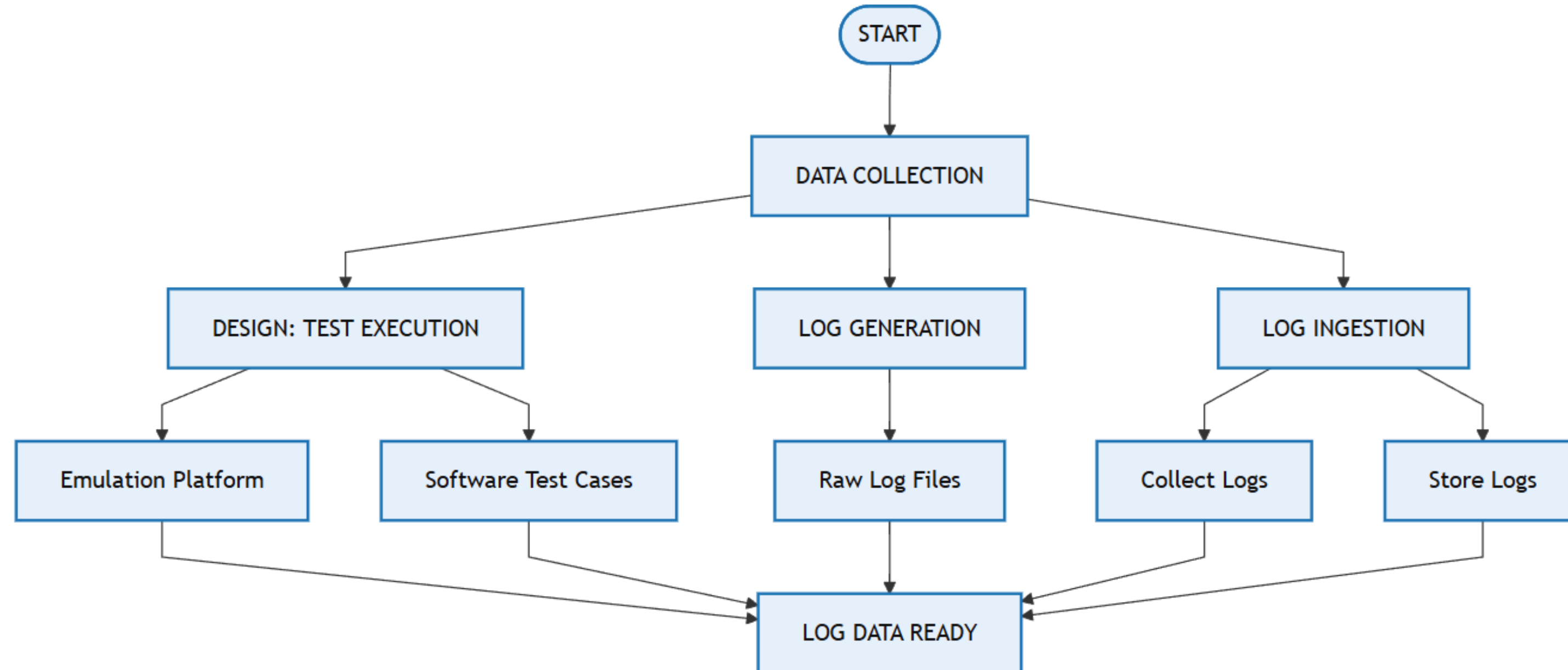
Literature Survey

Sl No	Paper Title, Journal Name, Year	Major findings / observations
5	Li, J. et al. (2025) – Practitioners' Expectations on Log Anomaly Detection. IEEE Transactions on Software Engineering.	Highlights the gap between academic research and real-world requirements in log anomaly detection systems.



Design Methodology - Data Collection

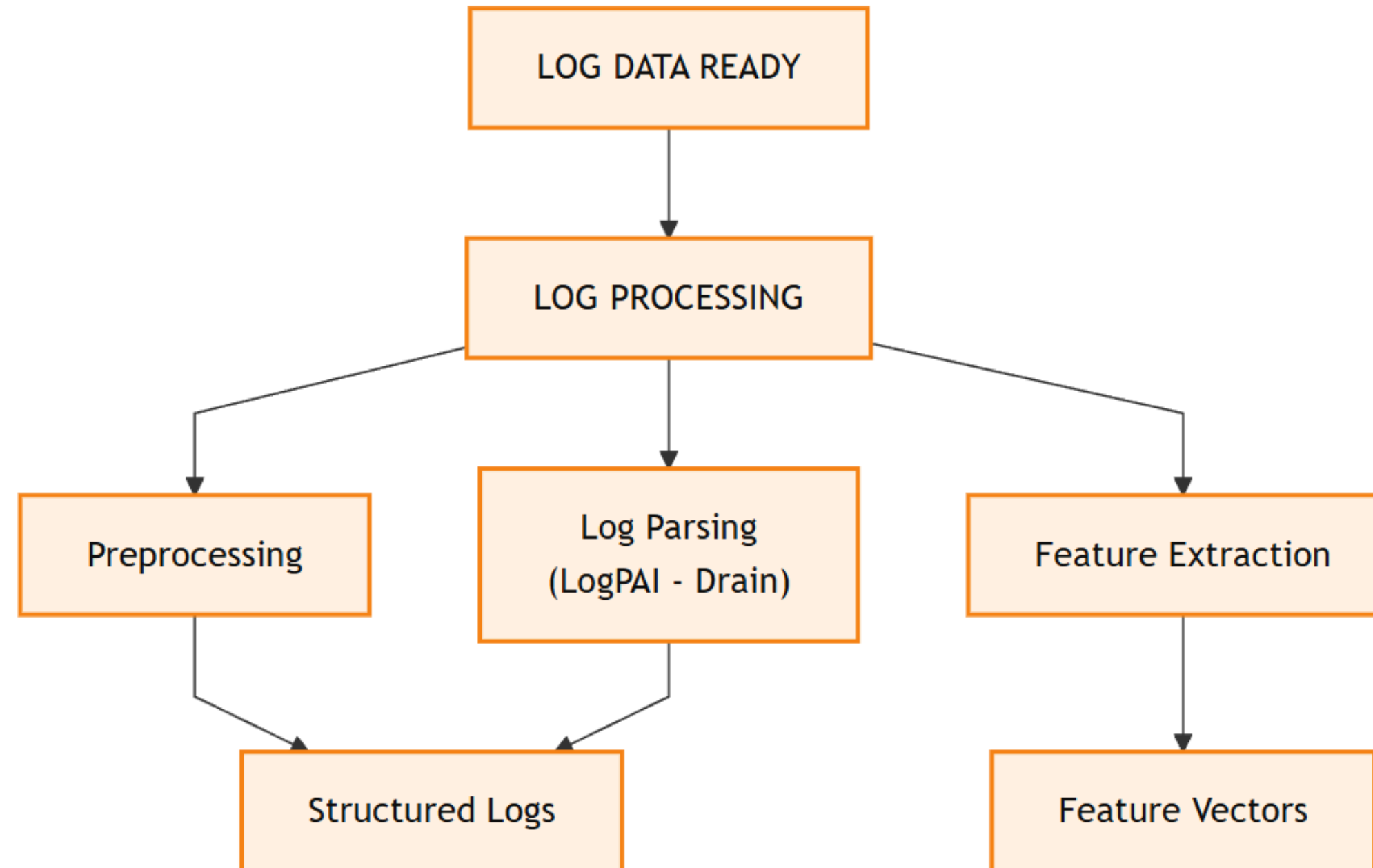
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Design Methodology-Log Processing

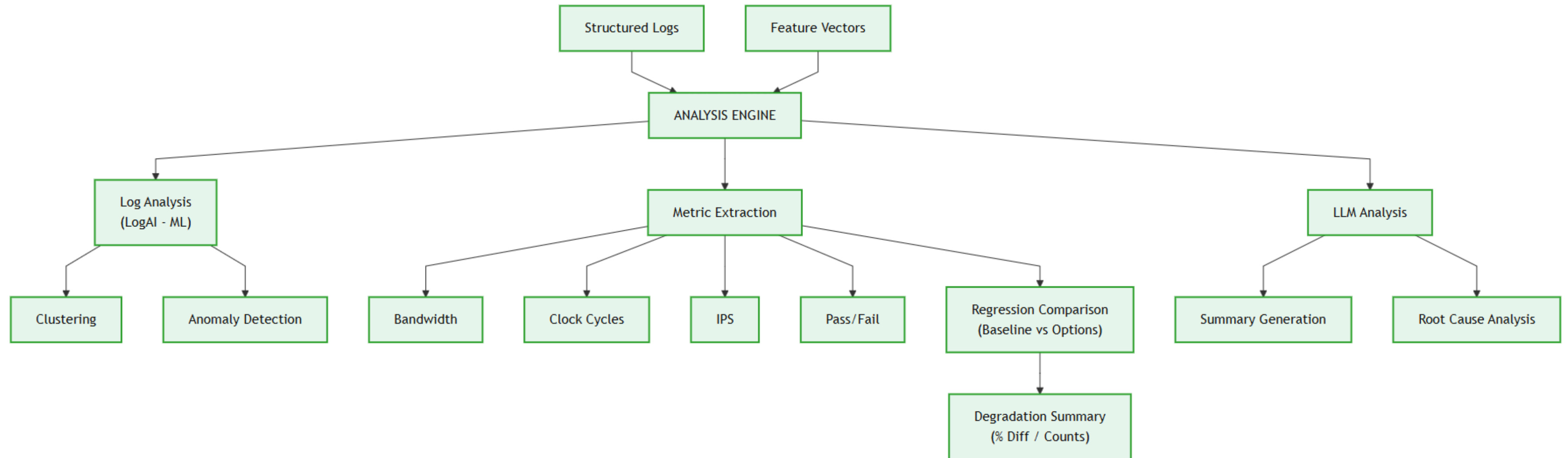
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Design Methodology-Analysis Engine

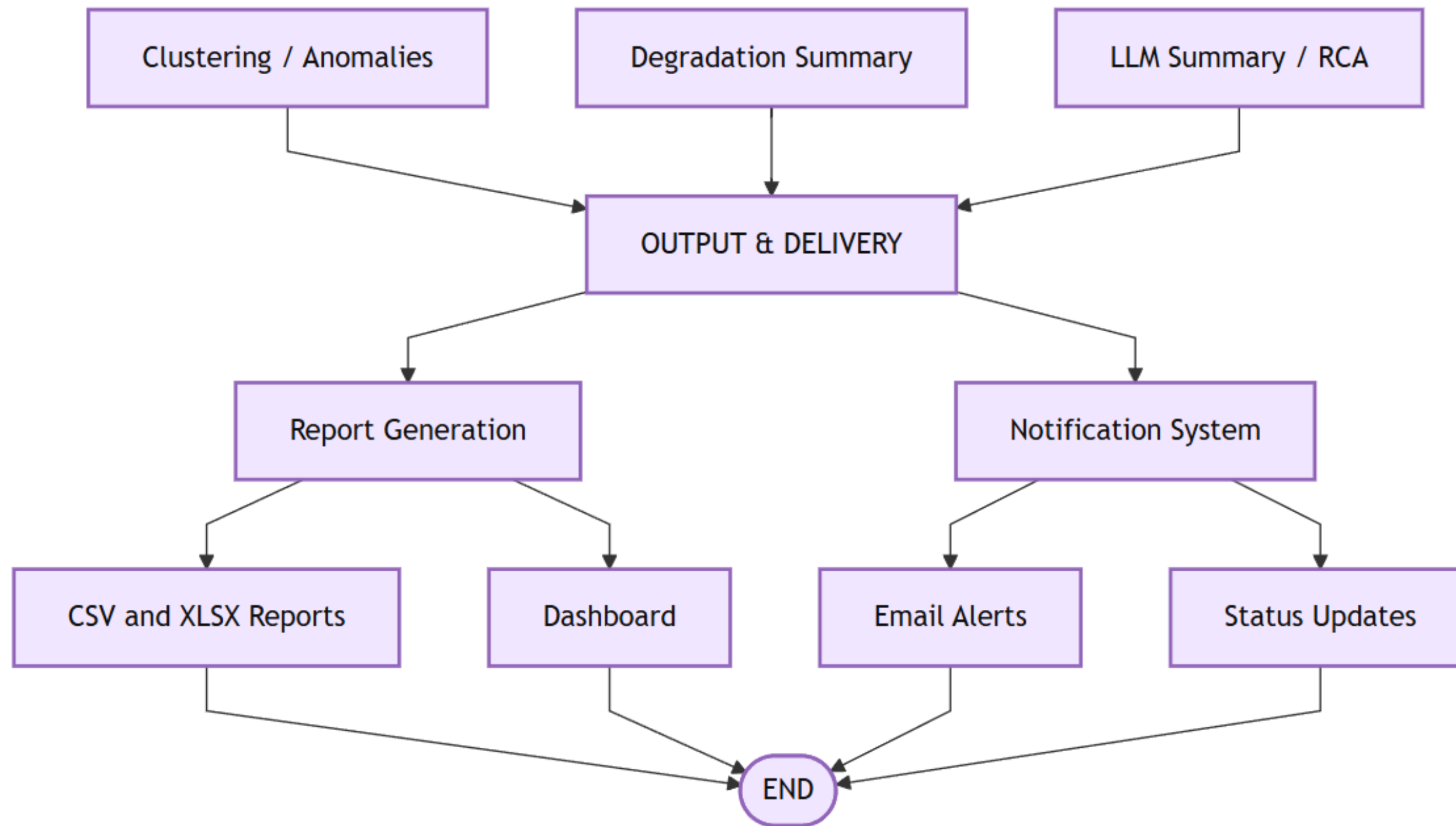
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Design Methodology-Output & Delivery

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Parameters

Parameter	Metric Group	What It Measures
cycles	Throughput	Total clock cycles per inference
IPS	Throughput	Inferences per second
bandwidth (MB)	Memory System	Total DDR read + write traffic
HMX_MAC / HMX_CVT	HMX Workload	HMX matrix / convert work packets
HVX_PKT	HVX Workload	HVX vector work packets
HMX_STALLS	Stall Analysis	HMX idle cycles (not doing useful work)
HVX_STALLS	Stall Analysis	HVX idle cycles
SCALAR	Scalar Activity	Aggregate scalar-core activity (PVIEW counter sum)



Parameters

Parameter	Metric Group	What It Measures
selected_sequencer	Scheduling	Compiler-chosen execution schedule
spill_bytes / fill_bytes	Memory Pressure	VTCM overflow traffic to DDR
hmx_parallelism / hvx_parallelism	multi_calc	Average parallelism factor
tiling_shape	Tiling	Most frequent tile shape used
cp_gt5_optypes	Ops	Ops contributing >5% of inference time
hexnn_sha	Build/Version	Compiler version identifier (commit)
booter_args	HW Configuration	Hardware register/config settings used
compilation_flags	Build Configuration	SW override flags used during compile



Test Suite (12 Graphs)

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Graph Name	Graph Type	Key Characteristic
MobileVision	coproc_1	Low BW, high IPS
MobileVision	coproc_2	Very high IPS
LLaMA2_1B_Prefill	coproc_1	LLM prefill, high BW
LLaMA2_1B_Decode	OTHER_Heavy	LLM decode, DMA-bound
LLaMA2_3B_Prefill	coproc_2	3B LLM prefill, very high BW
LLaMA2_3B_Decode	OTHER_Heavy	3B LLM decode, DMA-bound
VideoDenoise_HD	OTHER_Heavy	High spill, DMA-bound
MobileBERT	coproc_2	NLP transformer
MobileDetSSD	coproc_2	Real-time object detection
MosaicNet	coproc_2	Vision segmentation
EDSR_SuperRes	coproc_2	Super-resolution, high spill
MobileNetV4	coproc_1	Image classification



Details of Software / Hardware

- Python 3
- Pandas, NumPy
- Regular Expressions
- LLM API (GPT / Claude)
- CSV / Excel tools
- Git
- Linux (Ubuntu / RHEL)
- Emulation platform



Degradation Summary Results

Option(cores)	Total Tests	Pass	Fail	Degraded
Cascade (baseline)	12	12	0	—
Nova-Alpha	12	12	0	7
Nova-Beta	12	12	0	2
Nova-Gamma	12	12	0	3
Nova-Delta	12	12	0	1
Nova-Epsilon	12	12	0	1



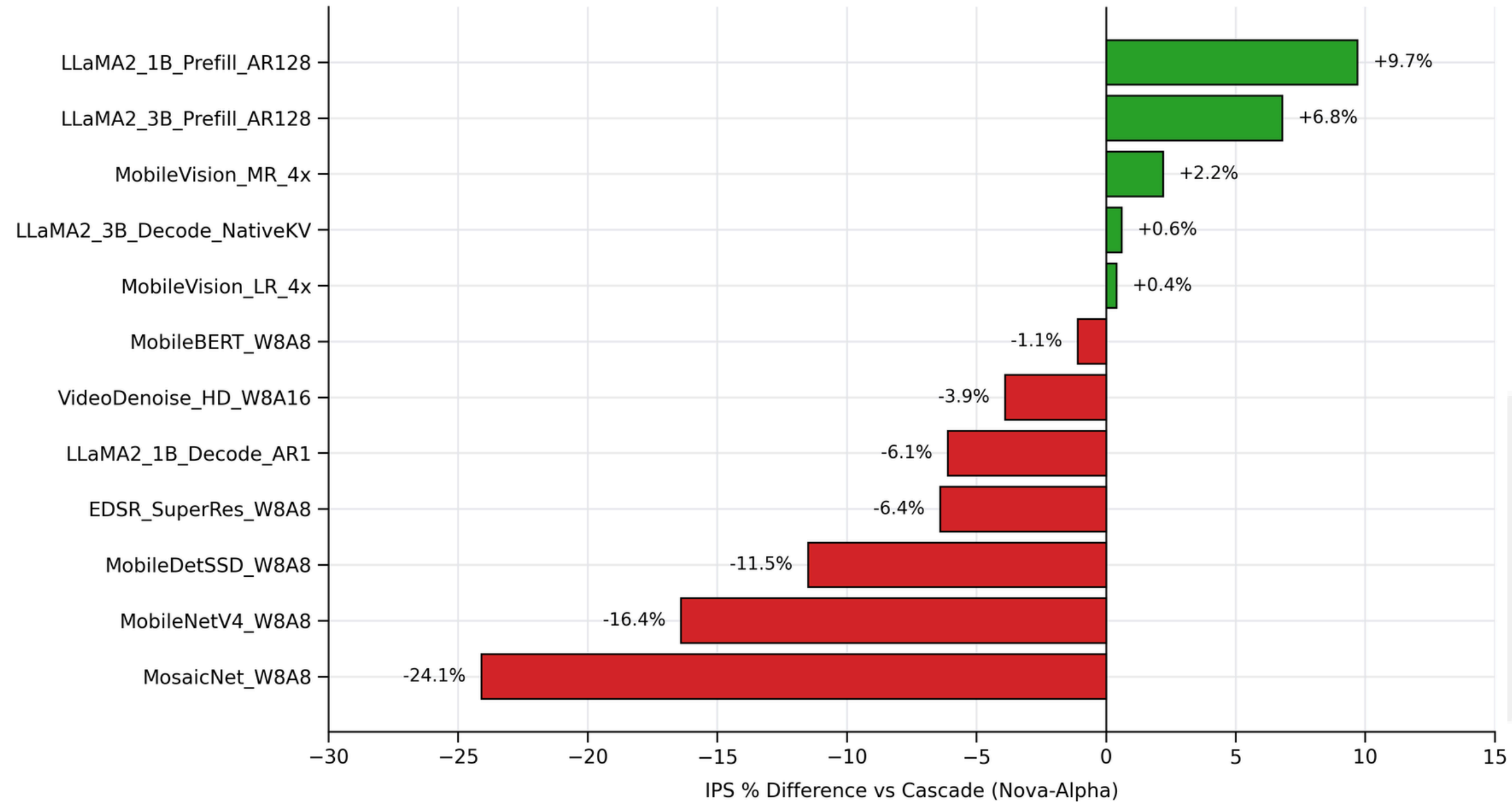
Degradation Summary Results

- Nova-Alpha (same clock as Cascade) has the most degradations — **7 out of 12 tests**
- Nova-Delta and Nova-Epsilon (1037 MHz) have only **1 degradation each** — best result
- Higher clock frequency compensates for most architectural differences
- All regressions passed (0 failures) — only performance degradations observed



IPS Results: Nova-Alpha vs Cascade (Same Clock)

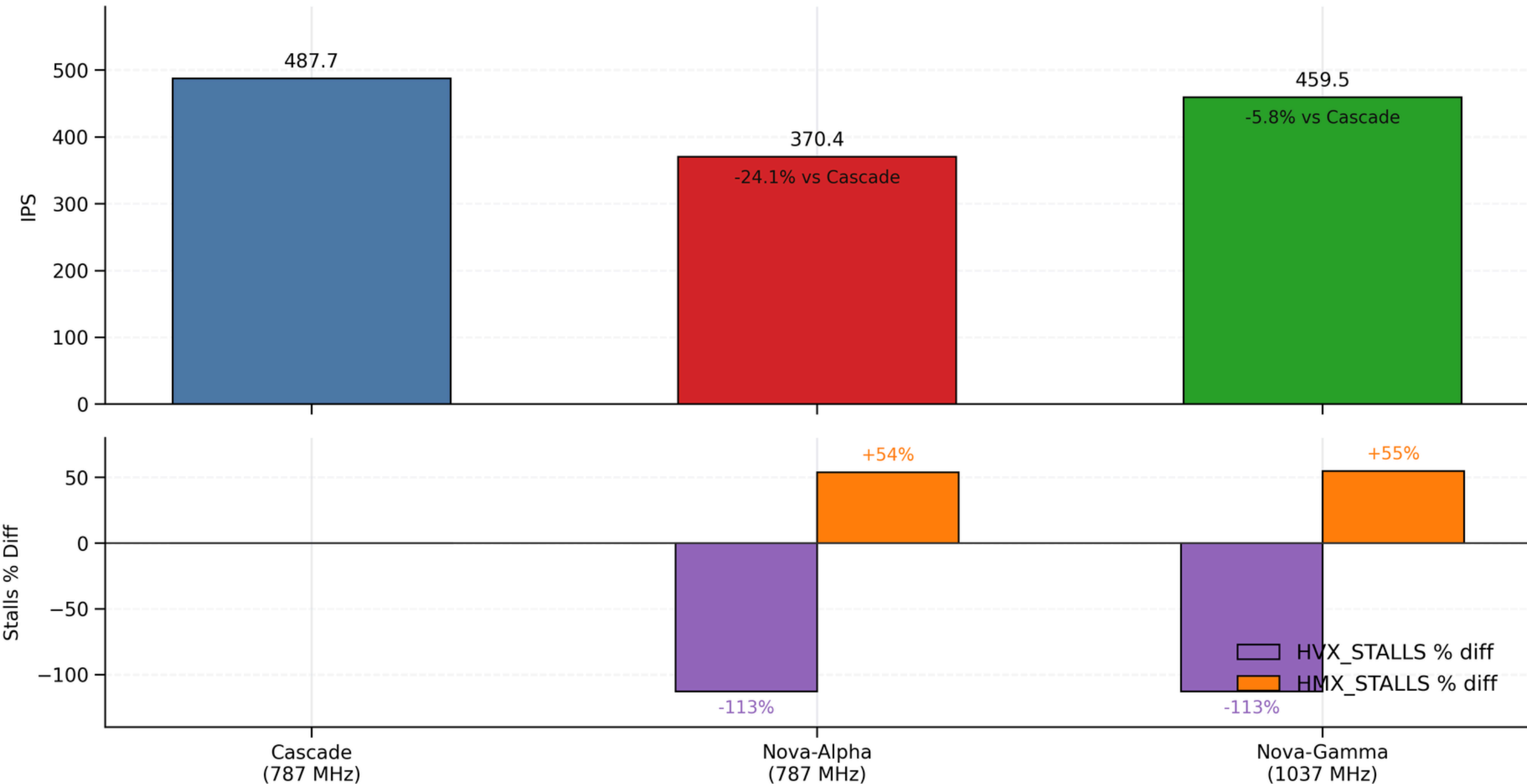
Nova-Alpha vs Cascade — IPS % Difference per Graph





Root Cause Analysis: MosaicNet_W8A8

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Analysis

- At 787 MHz (Nova-Alpha): HVX stalls explode (–113%) — HVX is severely starved
- At 1037 MHz (Nova-Gamma): HVX stalls are the same but higher clock compensates
- IPS recovers from –24.1% to –5.8%
- HMX stalls actually **improve** on Nova (fewer HMX stall cycles) — HMX is not the bottleneck
- Remaining –5.8% at 1037 MHz is due to HVX stalls that the higher clock cannot fully compensate



Workdone so far

- Implemented log ingestion and preprocessing of raw log files
- Developed basic log parsing using Python and regular expressions
- Extracted key performance metrics such as bandwidth, clock cycles, IPS, and pass/fail status
- Generated structured CSV reports using Pandas for analysis
- Integrated initial LLM-based module for basic log summary and insights



Work Plan for Next Phase

- Implement structured log parsing using LogPAI (Drain algorithm)
- Improve log parsing accuracy and handle different log formats
- Introduce LogAI-based analysis for basic clustering and anomaly detection
- Enhance LLM module for better summaries and root cause analysis
- Compare normal vs abnormal logs for performance insights
- Validate system using multiple log datasets



Timeline

	April 2025				May 2025				June 2025			
	Week 01	Week 02	Week 03	Week 04	Week 01	Week 02	Week 03	Week 04	Week 01	Week 02	Week 03	Week 04
Literature Survey	✓											
Specifications study	✓	✓	✓	✓								
Architecture study		✓	✓	✓	✓	✓	✓					
Design			✓	✓			✓	✓				
Development or Implementation				✓	✓	✓		✓	✓			
Software study			✓	✓	✓	✓	✓	✓				
Supporting courses	✓	✓	✓	✓	✓							
Documentation							✓	✓	✓	✓	✓	✓



References

SI No	Author, Title of paper, Journal	Year
1	Zhang, Y. et al. – SemanticLog: Towards Effective and Efficient Large-Scale Semantic Log Parsing. IEEE.	2025
2	Liu, X. et al. – LUK: Empowering Log Understanding With Expert Knowledge From Large Language Models. IEEE Transactions on Software Engineering.	2025
3	Wang, H. et al. – LogDLR: Unsupervised Cross-System Log Anomaly Detection Through Domain-Invariant Latent Representation. IEEE.	2025
4	Chen, Z. et al. – EPAS: Efficient Online Log Parsing via Asynchronous Scheduling. IEEE ICDE	2025
5	Li, J. et al. – Practitioners' Expectations on Log Anomaly Detection. IEEE Transactions on Software Engineering.	2025



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Thank You